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UNIVERSITÄT  
JENA**

INTERNSHIP PRESENTATION

**FRIEDRICH SCHILLER UNIVERSITY JENA**

**FACULTY OF PHYSICS AND ASTRONOMY**

**INSTITUTE FOR SOLID STATE THEORY AND OPTICS**

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# **Effect of Localized Rotation Errors on the Deutsch–Jozsa Algorithm**

Submitted in partial fulfillment of the requirements for the Internship Module

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# Why error location matters

## Research problem

- Quantum algorithms depend on precise gates and phase-sensitive interference.
- A coherent over-rotation can accumulate systematically instead of averaging away.
- The same rotation may be harmless at one stage but destructive at another.

## Study focus

- Compare localized  $R_x(\theta)$ ,  $R_y(\theta)$ , and  $R_z(\theta)$  errors.
- Test four physically meaningful locations,  $E_1$ – $E_4$ .
- Identify the axis–location combinations that most reduce  $P(0^n)$ .

# The study separates axis, angle, and location

**1 Model** Implement coherent single-qubit rotations  $R_x(\theta)$ ,  $R_y(\theta)$ , and  $R_z(\theta)$ .

**2 Localize** Insert each error separately at  $E_1$ ,  $E_2$ ,  $E_3$ , and  $E_4$ .

**3 Measure** Sweep  $\theta$  and evaluate the probability  $P(0^n)$ .

**4 Compare** Identify robust and vulnerable stages of the algorithm.

# One promise problem reveals a query advantage

$$f : \{0,1\}^n \rightarrow \{0,1\}$$

Promise

**Constant**

$f(x)$  has the same value  
for every input  $x$ .

**Balanced**

$f(x) = 0$  for half the inputs  
and  $f(x) = 1$  for the other half.

**Task**

Determine with certainty whether the promised function is constant or balanced.

# Quantum interference compresses many function values into one query

To decide with certainty

Classical deterministic(worst case)

$$2^{n-1} + 1$$

oracle queries in the worst case

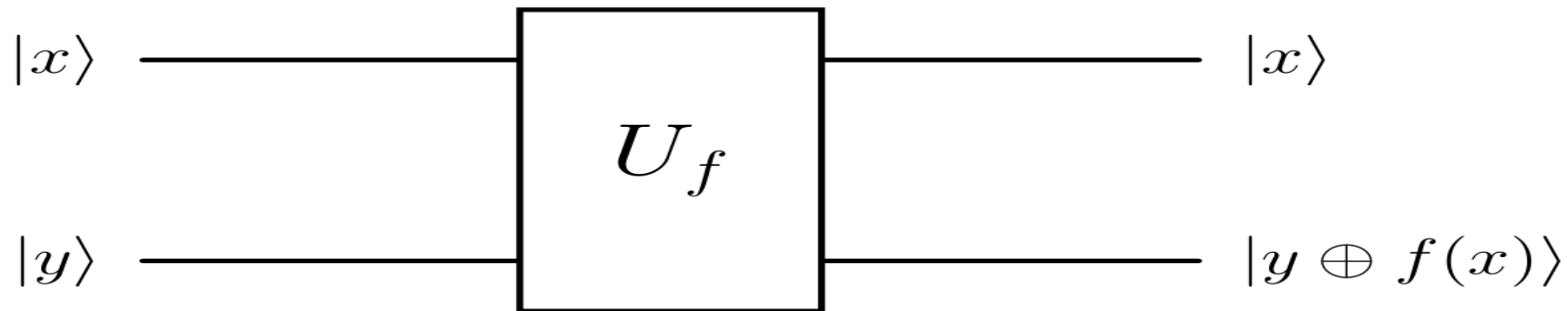
Deutsch–Jozsa quantum algorithm

**1** oracle query

The advantage comes from superposition and interference—not from reading every  $f(x)$ .

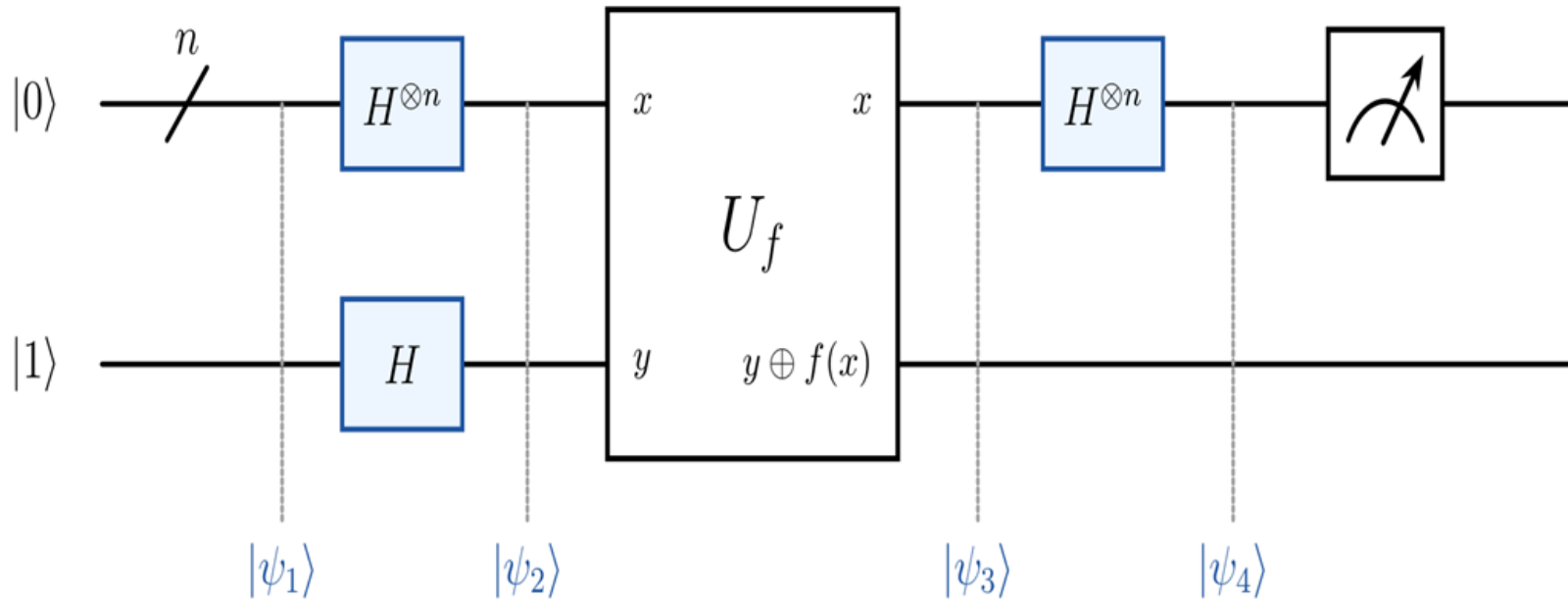
# The oracle stores $f(x)$ reversibly

$$U_f : |x\rangle|y\rangle \longmapsto |x\rangle|y \oplus f(x)\rangle$$



$x$  is unchanged; the ancilla changes from  $y$  to  $y \oplus f(x)$ . One passage through  $U_f$  is one oracle query.

# The ideal circuit converts phase structure into a binary decision



## Decision rule

Constant

$$P(0^n) = 1$$

Balanced

$$P(0^n) = 0$$

## Circuit logic

Superposition  $\rightarrow$  oracle phase encoding  $\rightarrow$  interference  $\rightarrow$  measurement

# Interference creates the all-zero signature

1

## Prepare

$|0\rangle^n|1\rangle$

2

## Superpose

H creates all  $2^n$  inputs

3

## Encode

oracle adds phases  $(-1)^{f(x)}$

4

## Interfere

final H combines  
amplitudes

### Output

Constructive interference at  $|0^n\rangle$  signals constant; cancellation at  $|0^n\rangle$  signals balanced.



# Three coherent rotations model axis-specific gate miscalibration

$$R_x(\theta)$$

$$\exp(-i\theta X/2)$$

Bit-axis rotation

$$R_y(\theta)$$

$$\exp(-i\theta Y/2)$$

Mixed bit-phase rotation

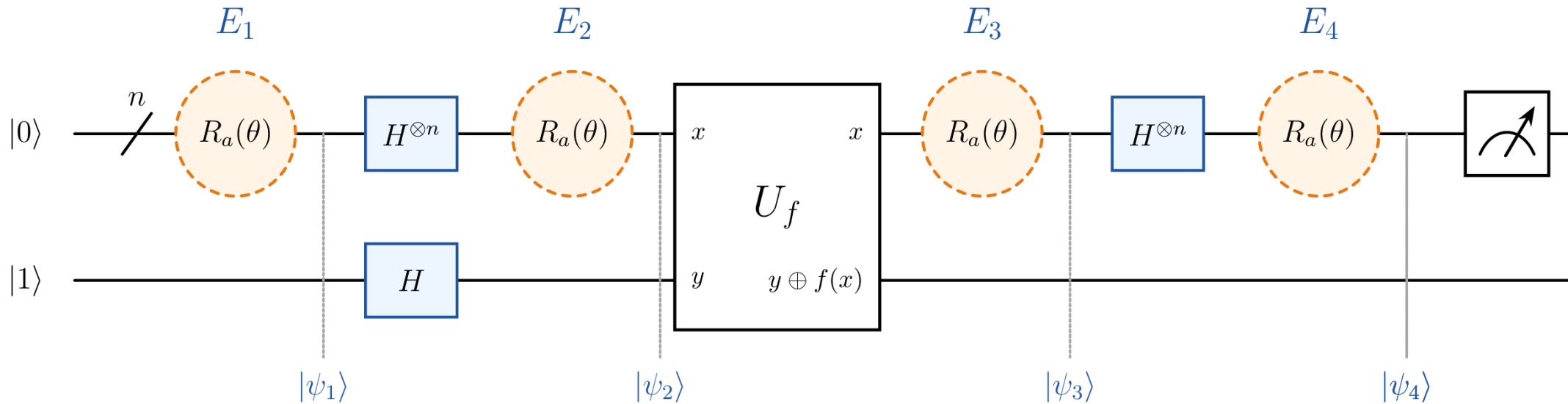
$$R_z(\theta)$$

$$\exp(-i\theta Z/2)$$

Phase-axis rotation

**Coherent error** A fixed unitary over- or under-rotation;  $\theta$  controls its magnitude and the axis controls its mechanism.

# Four locations isolate when the same rotation acts



- $E_1$ : Before the first Hadamard layer
- $E_2$ : After the first Hadamard layer and before  $U_f$
- $E_3$ : After  $U_f$  and before the final Hadamard layer
- $E_4$ : After the final Hadamard layer and before measurement

**$E_1$**  State preparation

**$E_2$**  Oracle input

**$E_3$**  Oracle output

**$E_4$**  Before measurement

# A controlled sweep holds everything else fixed

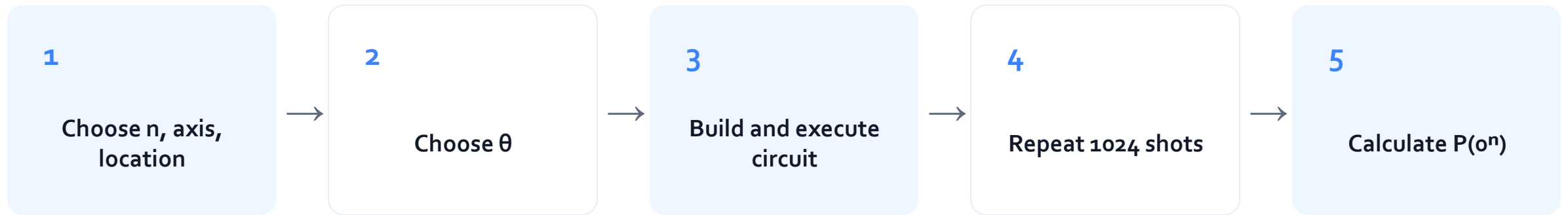
## Error implementation

- Apply one localized rotation to one selected data qubit.
- Test  $R_x(\theta)$ ,  $R_y(\theta)$ , and  $R_z(\theta)$  separately.
- Repeat independently at  $E_1$ ,  $E_2$ ,  $E_3$ , and  $E_4$ .

## Simulation parameters

- Input qubits:  $n \in \{1, 3, 5, 7, 9\}$ ; target  $q = \lfloor n/2 \rfloor$ .
- Angle:  $0^\circ$ – $180^\circ$  in  $20^\circ$  increments.
- 1024 measurement shots per circuit execution.

# Every trial produces one comparable success probability



Performance metric

$$P(o^n) = N(o^n) / 1024$$

Fraction of shots that produce the all-zero data register

# Sensitivity is an axis–location interaction

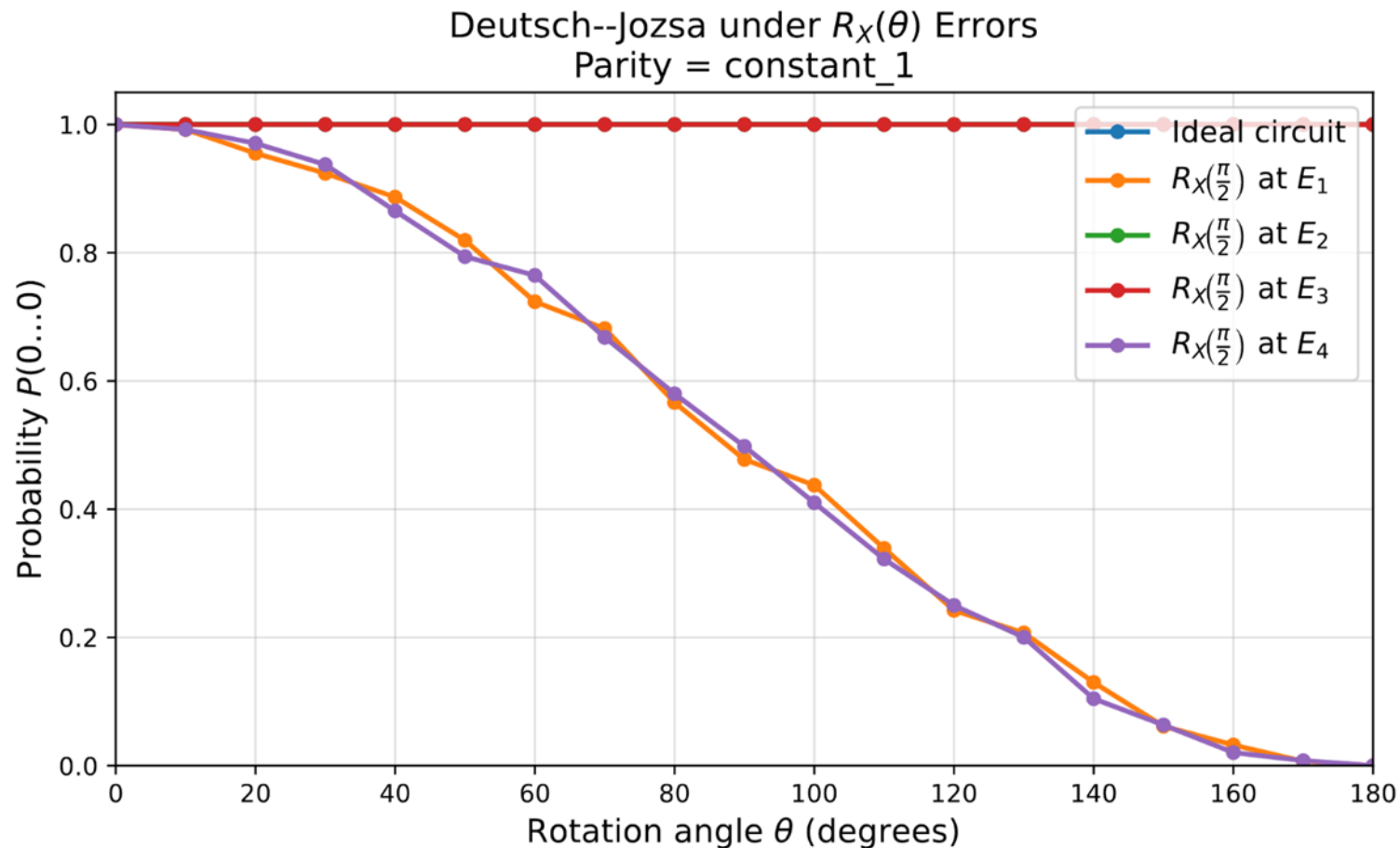
Nine plots test two constant functions and one balanced function across X, Y, and Z rotations.

**X** $E_1$  and  $E_4$ **Y**

All locations

**Z** $E_2$  and  $E_3$

# For constant $f(x)=1$ , X rotations expose $E_1$ and $E_4$



Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

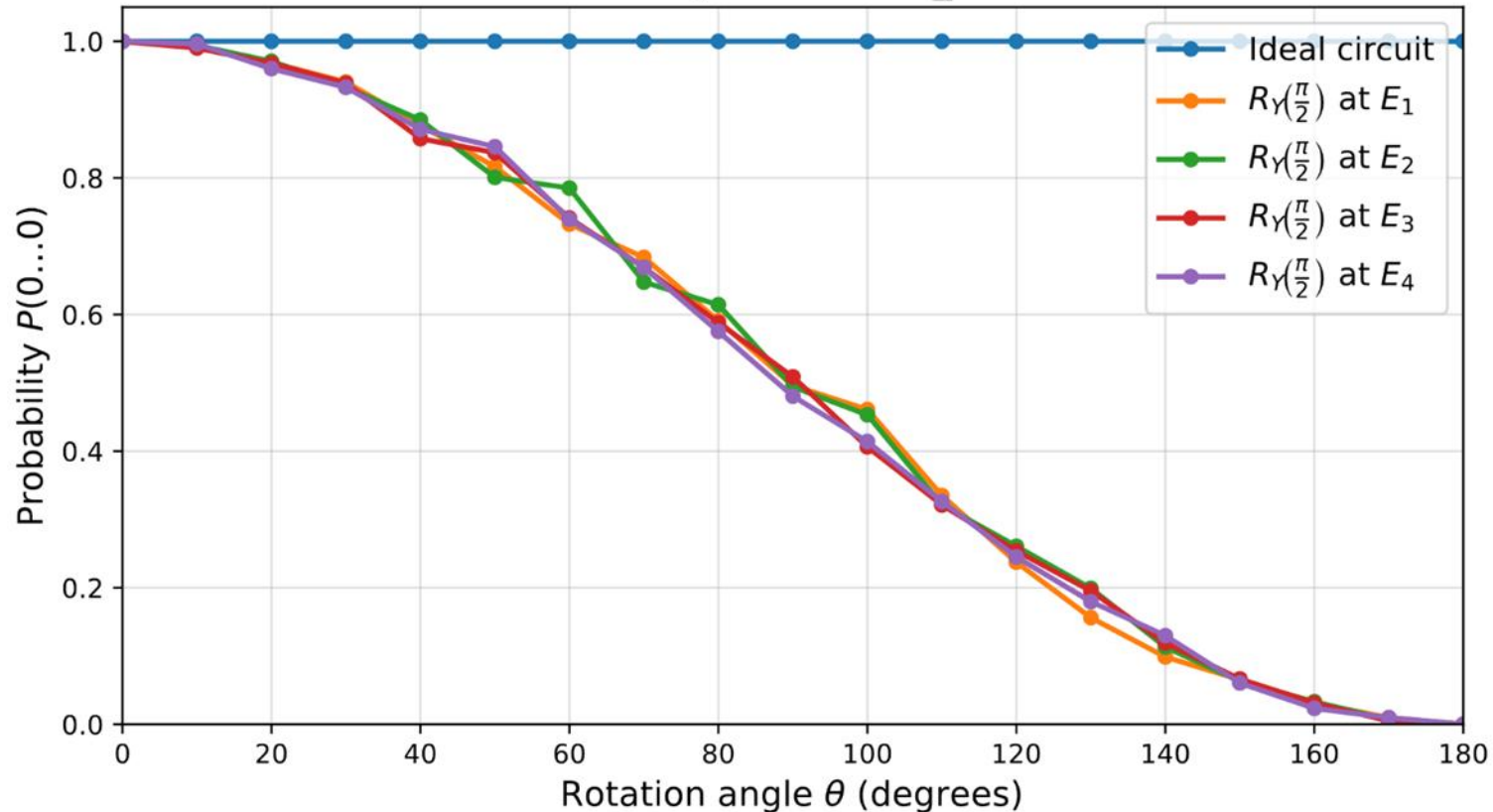
## X-axis rotation

- $E_1$  and  $E_4$  degrade together from  $P(0^n) \approx 1$  toward 0.
- $E_2$  and  $E_3$  remain close to the ideal curve.
- At intermediate stages, X-basis eigenstates turn the rotation mainly into phase.

**Most sensitive: state preparation and pre-measurement**

# For constant $f(x)=1$ , Y rotations affect every stage

Deutsch--Jozsa under  $R_Y(\theta)$  Errors  
Parity = constant\_1



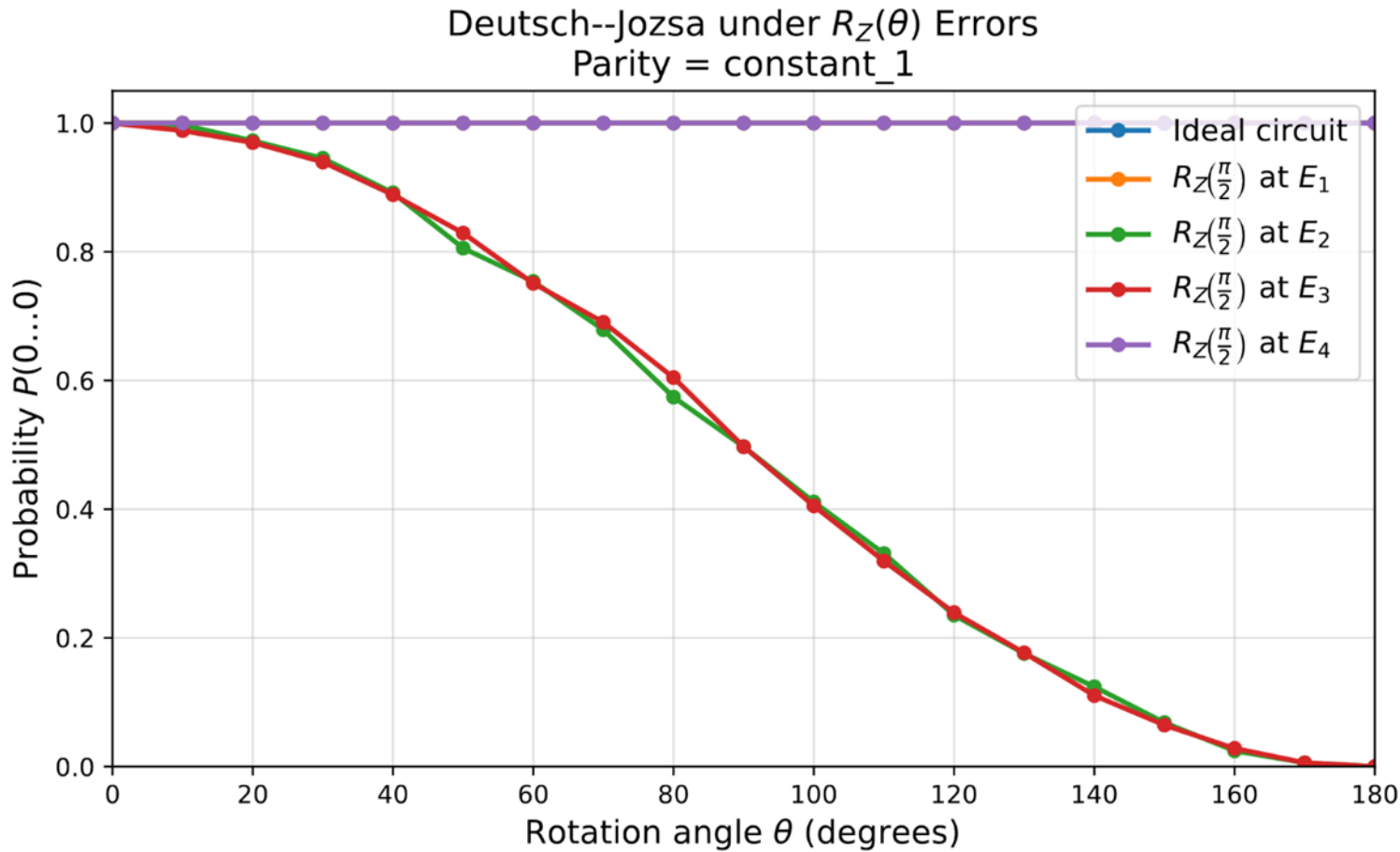
Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

## Y-axis rotation

- All four error-location curves nearly overlap.
- $P(0^n)$  decreases steadily as  $\theta$  approaches  $180^\circ$ .
- Rotation angle dominates; no location is substantially protected.

Uniform location sensitivity

# For constant $f(x)=1$ , Z rotations target $E_2$ and $E_3$



Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

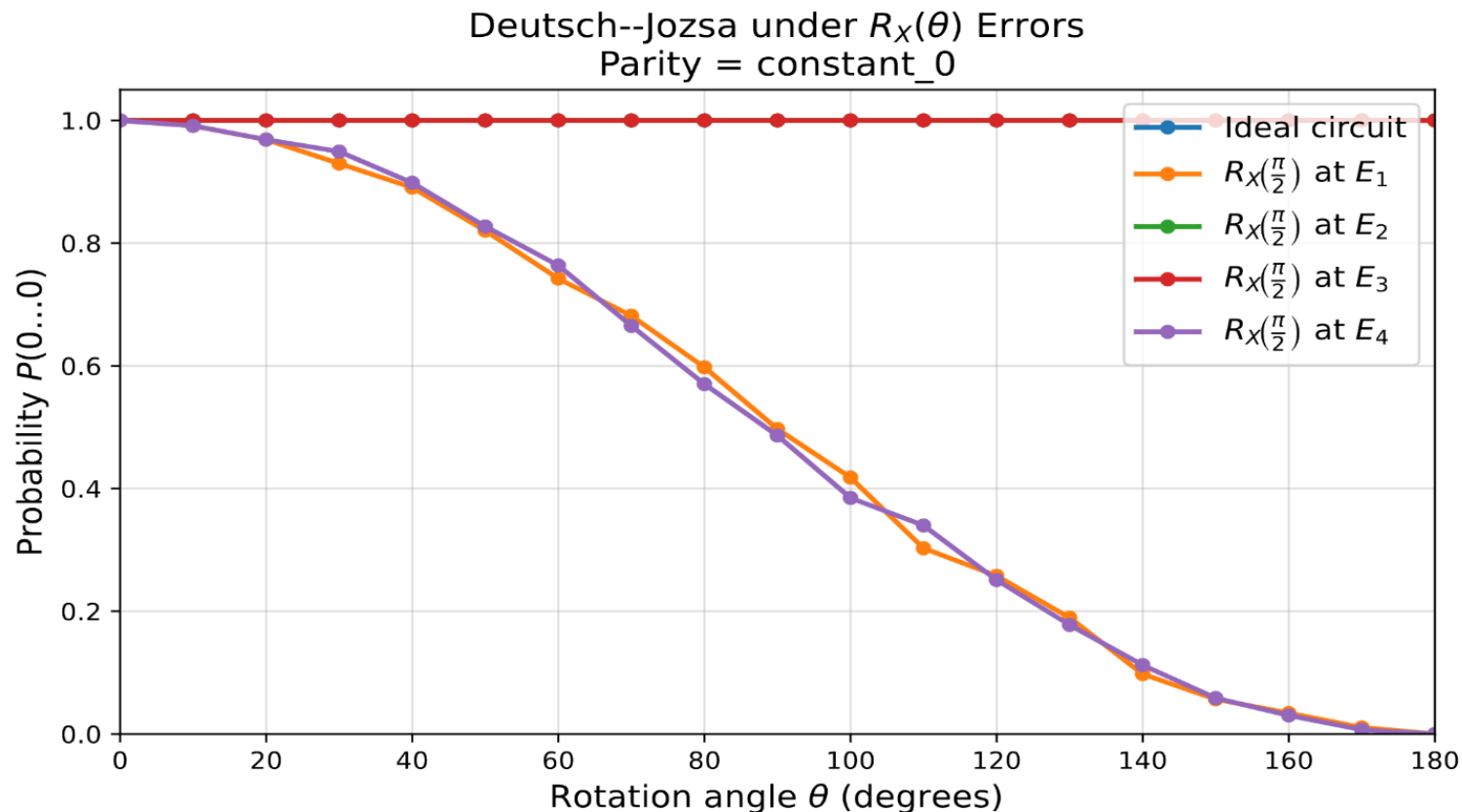
## Z-axis rotation

- $E_1$  and  $E_4$  remain essentially ideal.
- $E_2$  and  $E_3$  lose success probability as  $\theta$  increases.
- Relative phase changes inside the interference region alter the final outcome.

**Most sensitive: oracle input and oracle output**



# Constant $f(x)=0$ reproduces the X-axis boundary sensitivity



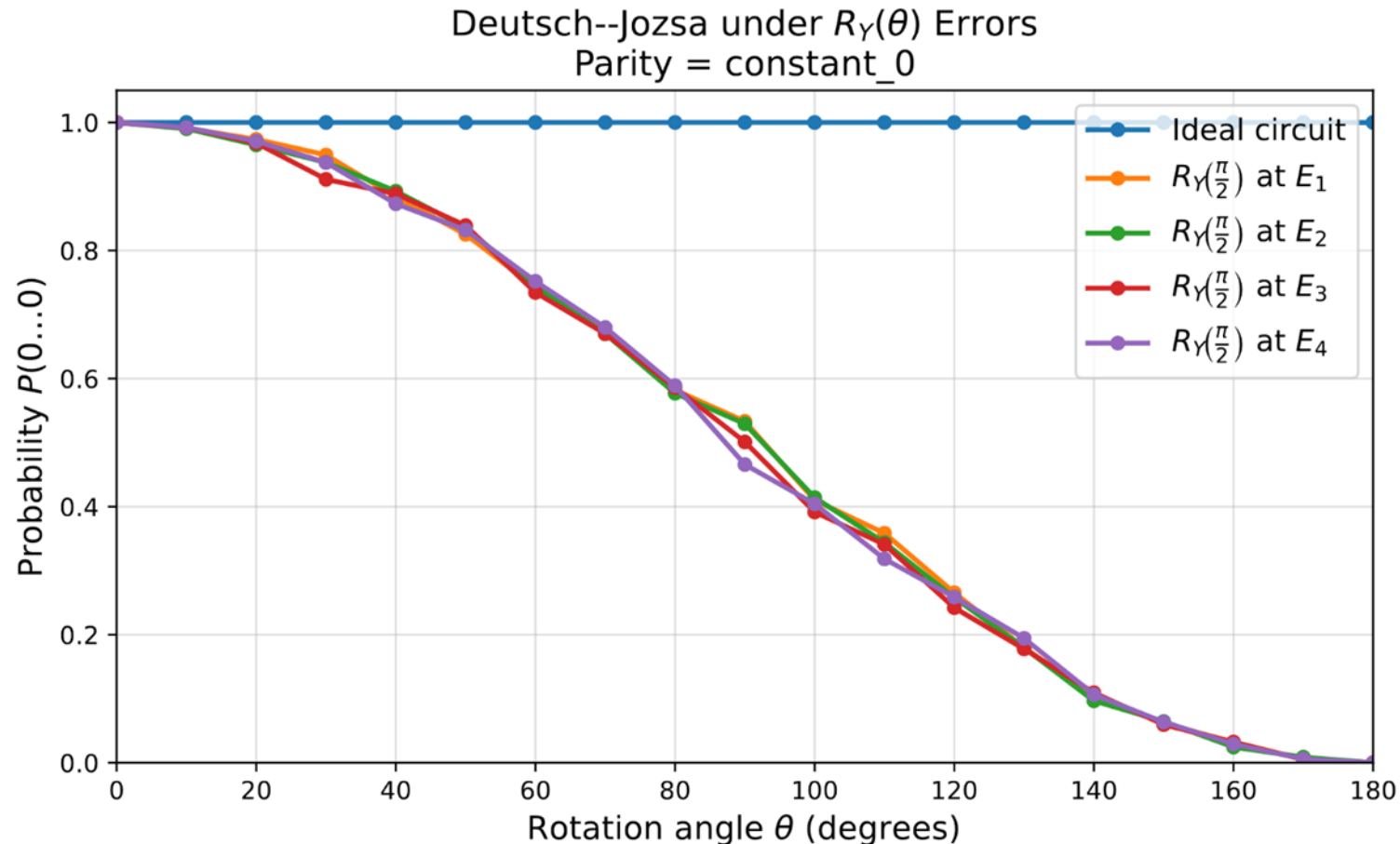
Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

## X-axis rotation · constant 0

- $E_1$  and  $E_4$  again show the dominant degradation.
- $E_2$  and  $E_3$  remain near the ideal result.
- The location pattern is not specific to the constant output value.

Robust pattern across both constant functions

# Constant $f(x)=0$ reproduces the uniform Y-axis response



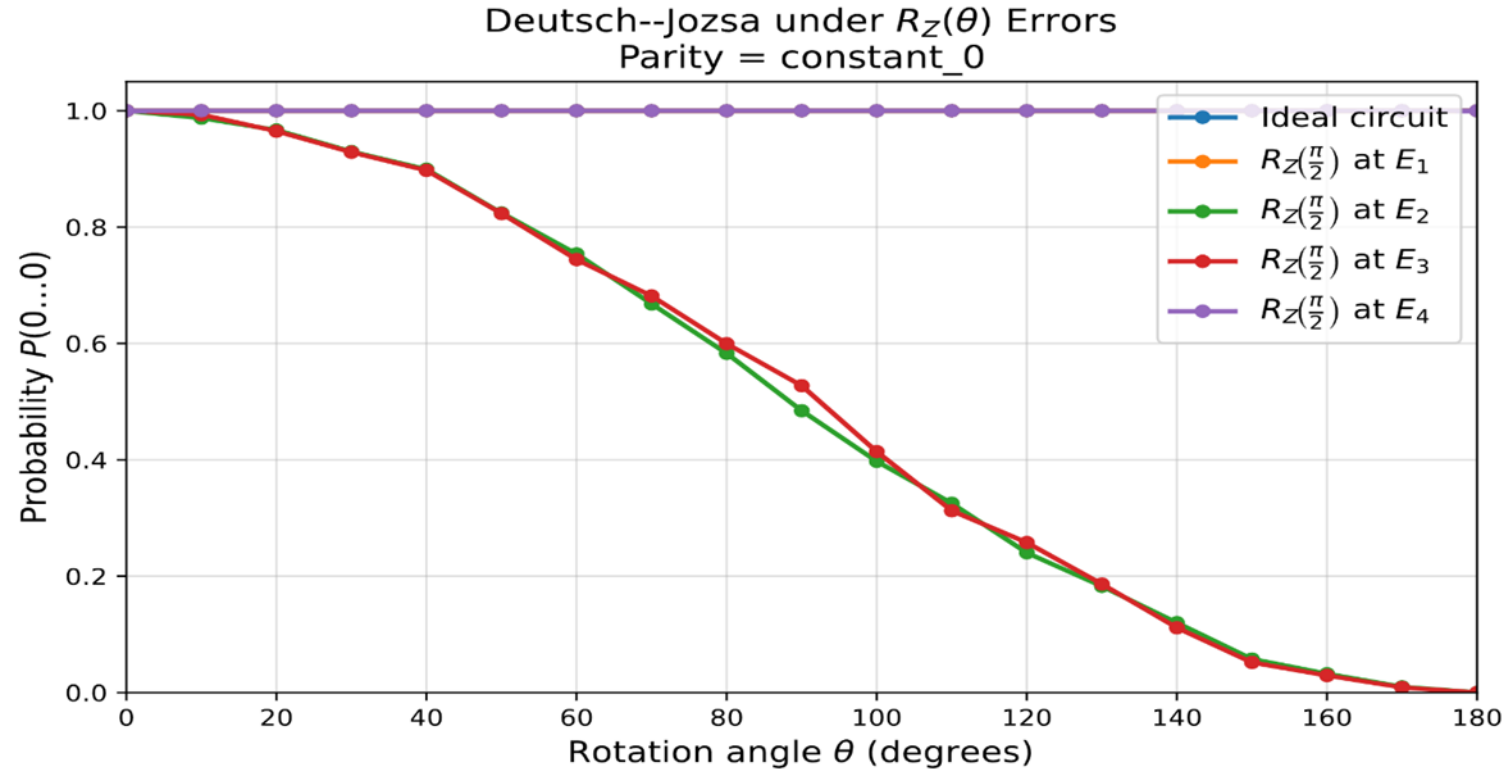
Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

## Y-axis rotation · constant 0

- All locations decrease along nearly the same trajectory.
- Differences are consistent with finite-shot fluctuations.
- Angle remains more important than position for Y rotations.

Y response is location-independent

# Constant $f(x)=0$ reproduces the Z-axis interference sensitivity

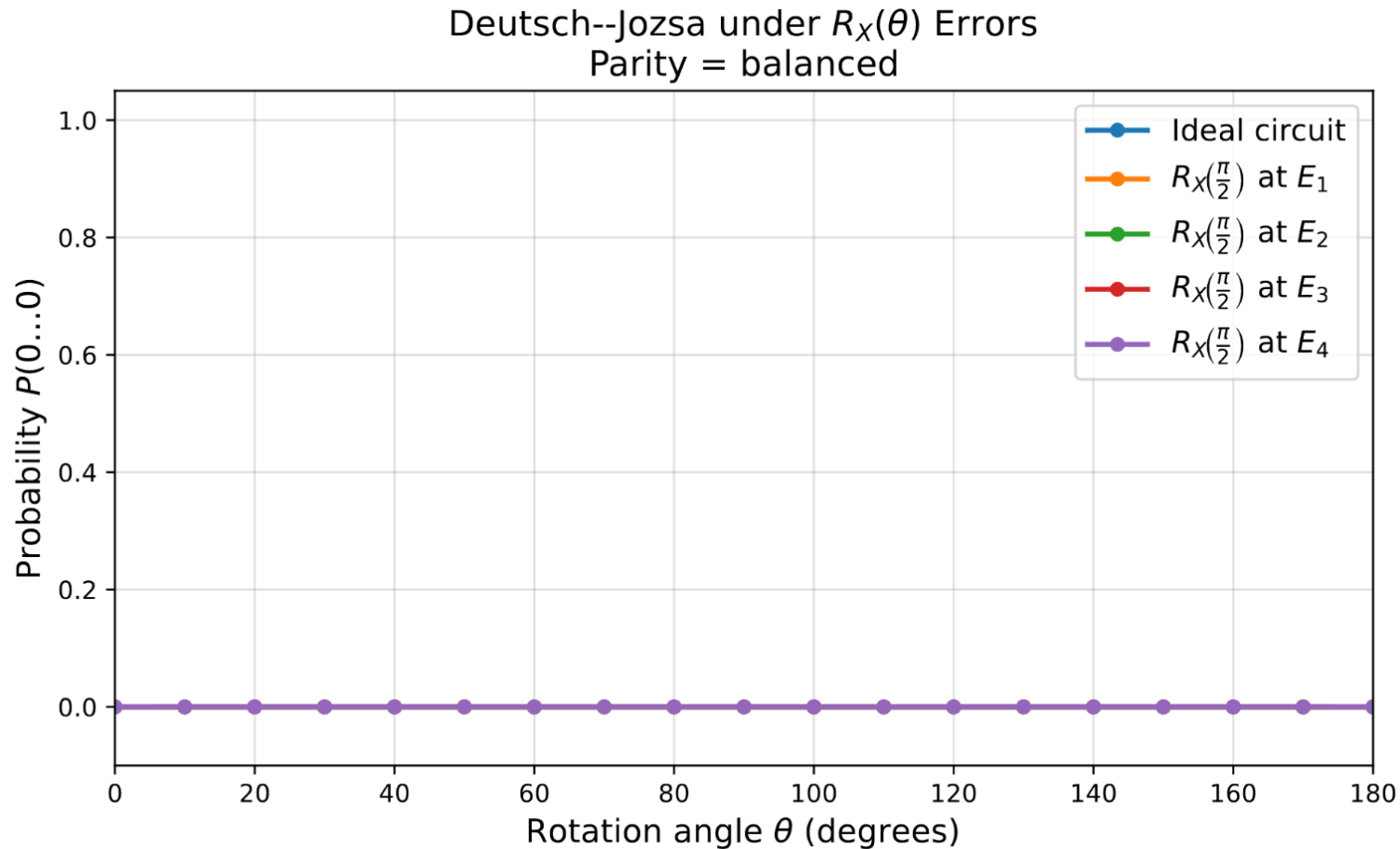


Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

## Z-axis rotation · constant 0

- $E_2$  and  $E_3$  are again the vulnerable positions.
- $E_1$  and  $E_4$  stay near  $P(0^n)=1$ .
- The same phase-sensitive mechanism holds for both constant oracles.

# Balanced parity remains classified under X rotations

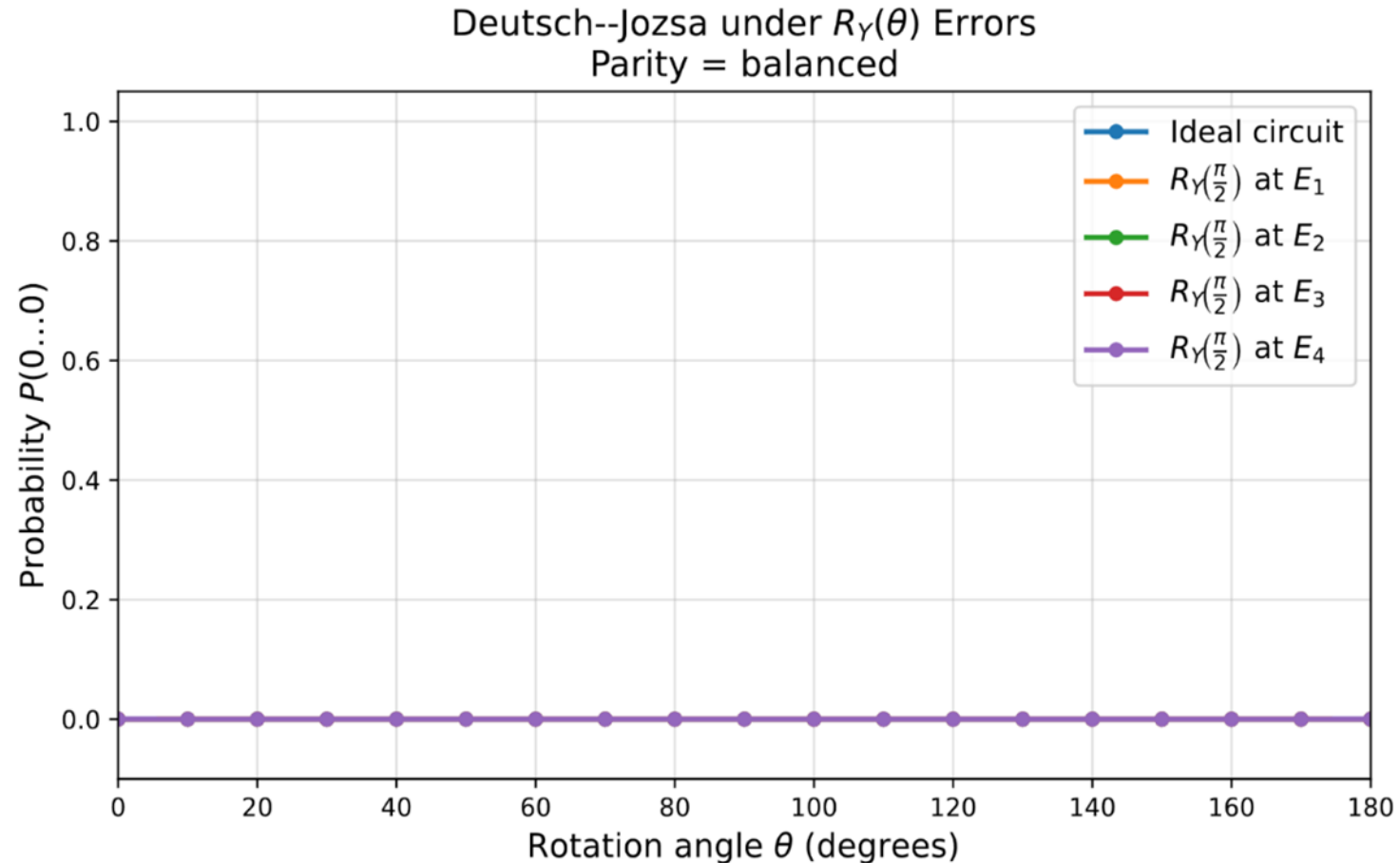


Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

## X-axis rotation · balanced

- $P(0^n)$  remains approximately zero for all locations and angles.
- All curves overlap with the ideal balanced result.
- This metric does not reveal changes among nonzero bit strings.

# Balanced parity remains classified under Y rotations

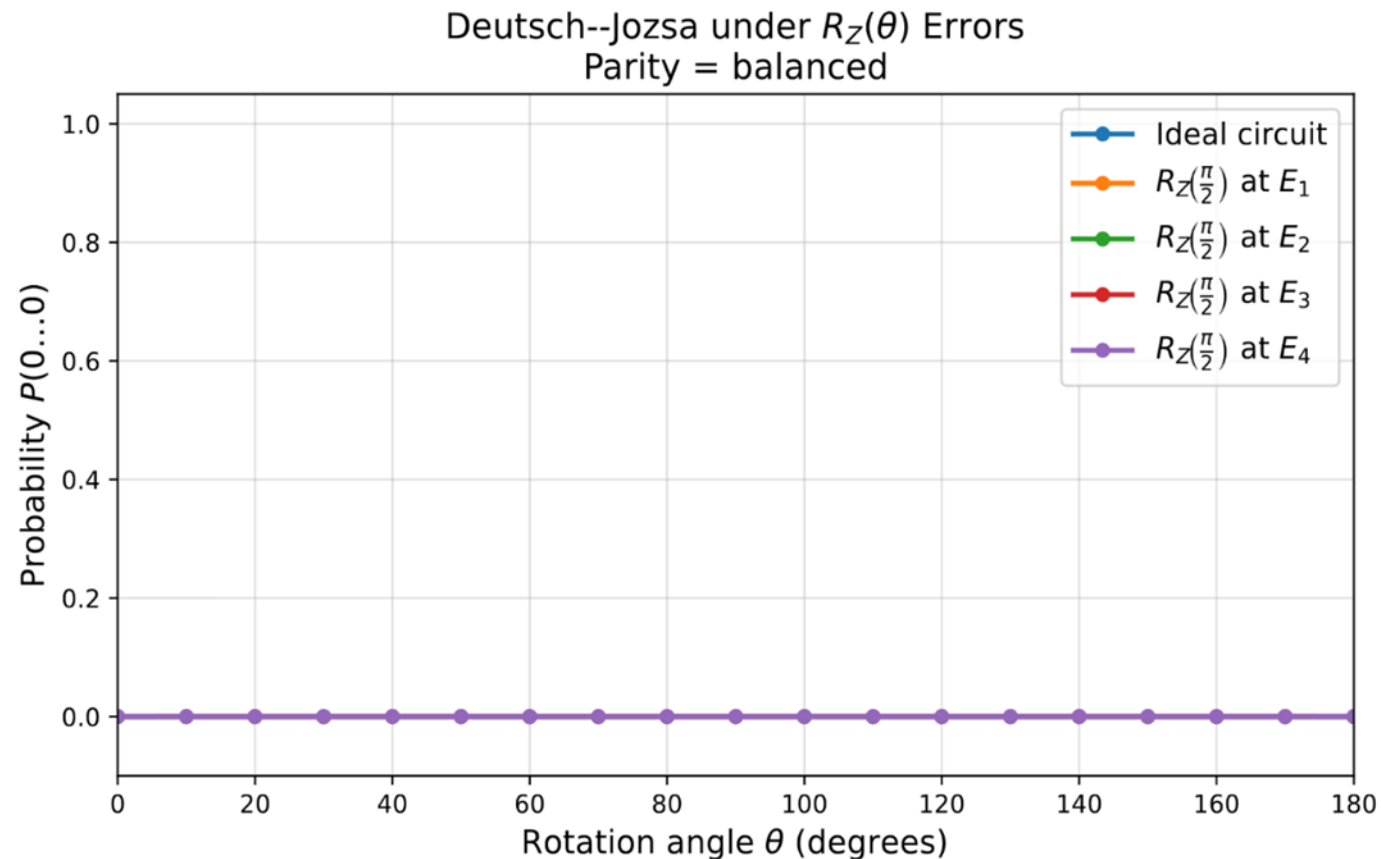


Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

## Y-axis rotation · balanced

- The all-zero probability stays at numerical zero.
- No axis–location dependence appears in  $P(0^n)$ .
- A full-distribution metric is needed to test whether other outputs changed.

# Z rotations preserve the balanced classification



Each plotted rotation uses the angle sweep  $\theta = 0^\circ \dots 180^\circ$

## Z-axis rotation · balanced

- $P(0^n)$  remains approximately zero throughout the sweep.
- The decision rule still reports balanced.
- Stable classification does not imply that the complete output state is unchanged.

# Each axis has a distinct vulnerability map

| Rotation      | $E_1$ | $E_2$ | $E_3$ | $E_4$ |
|---------------|-------|-------|-------|-------|
| $R_x(\theta)$ | HIGH  | LOW   | LOW   | HIGH  |
| $R_y(\theta)$ | HIGH  | HIGH  | HIGH  | HIGH  |
| $R_z(\theta)$ | LOW   | HIGH  | HIGH  | LOW   |

**Interpretation** X errors attack circuit boundaries; Z errors attack the interference region; Y errors affect all stages.

# The conclusions are precise

## What this study establishes

- Axis and circuit position jointly determine coherent-error sensitivity.
- The pattern repeats across both constant functions.
- Balanced classification is stable under the selected metric.

## What it does not claim

- No depolarizing, damping, decoherence, or stochastic-noise model.
- One selected input qubit; no correlated or multi-qubit errors.
- $P(0^n)$  does not expose redistribution among nonzero outcomes.

### Experimental scope

Simulation with 1024 shots; hardware drift and device-specific calibration are not modeled.



# Error mitigation should be matched to the error mechanism

**X**

Protect preparation and pre-measurement stages

$E_1$  and  $E_4$

**Y**

Treat the circuit as uniformly exposed

$E_1$ – $E_4$

**Z**

Protect the phase-sensitive interference region

$E_2$  and  $E_3$

# Thank You

- Any

- Questions?